A brick column cooled by air on one side

Step-by-step COMSOL Multiphysics 4.3a guide for students

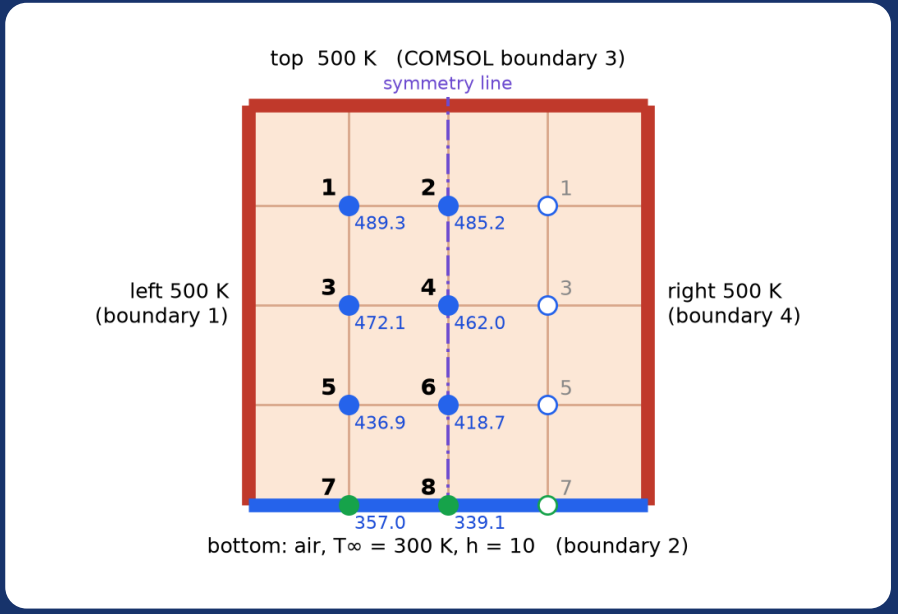
Problem: 2D steady conduction with a convection boundary, Dx = Dy = 0.25 m (Introduction to CFD, Problem 4; Incropera Example 4.3)

Tool: COMSOL Multiphysics 4.3a, Heat Transfer in Solids with Convective Cooling, mesh-size sweep

Result (finite difference, 8 unknowns): T = 489.3, 485.2, 472.1, 462.0, 436.9, 418.7, 357.0, 339.1 K; q' = 882.6 W/m

COMSOL, converged mesh: q' = 623.6 W/m (the 4 x 4 grid over-predicts the heat rate by 41 %)

Includes: hand solution (energy balances, Gauss, Gauss-Seidel), 16 annotated COMSOL steps, grid study

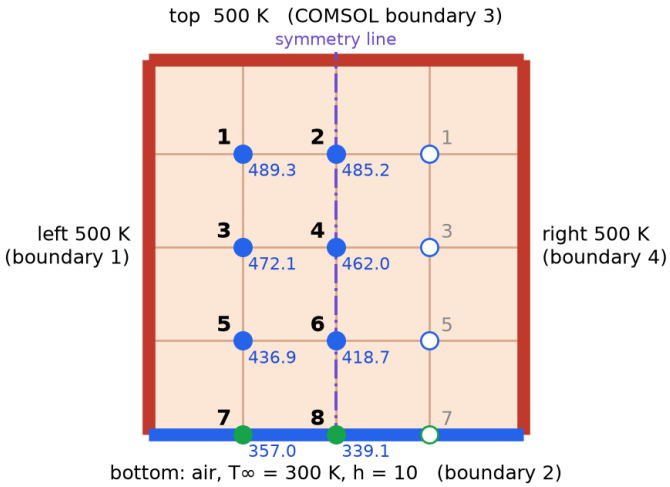


Model file: brick_column_comsol.mph (open it in COMSOL to compare with your own model)

The problem and the finite difference setup

A furnace is supported on a long column of fireclay brick, 1 m by 1 m. In steady state three surfaces of the column are at 500 K and the remaining surface is exposed to an air stream (T_inf = 300 K, h = 10 W/(m^2 K)). With a grid Dx = Dy = 0.25 m determine the 2D temperature distribution and the heat rate to the air stream per unit length of column. (Adapted from Incropera et al., Example 4.3.)

Quantity	Symbol	Value
width = height	W	1 m
grid spacing	Dx = Dy	0.25 m
conductivity (brick)	k	1 W/(m K)
hot surfaces (3)	T_s	500 K
air temperature	T_inf	300 K
convection coefficient	h	10 W/(m^2 K)



Assumptions: steady state; 2D conduction; no heat generation; constant k and h; radiation neglected.

1. Physics

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Top, left, right: T = 500 K. Bottom (convection): -k dT/dy = h (T - T_inf), with the Biot number of a grid cell Bi = h Dx / k = 10 x 0.25 / 1 = 2.5.

2. Grid and symmetry

The temperature is symmetric about the vertical centre line x = 0.5, so T(0.75, y) = T(0.25, y). The unknowns are the 8 nodes with x = 0.25 and 0.5 (nodes 1 to 8, figure). The nodes on the hot edges are known (500 K). Nodes 7 and 8 lie on the convective surface.

3. Node equations (energy balance, Dx = Dy)

Interior node: T = average of the four neighbours.
Node on the symmetry line: the neighbour across the line is the mirror image, so the left neighbour counts twice.

Convective surface node (half control volume, heat balance):

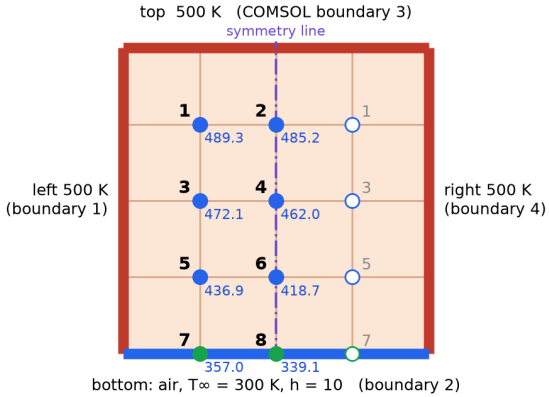
$$2T_{below} + T_{left} + T_{right} + 2\,Bi\,T_\infty - 2\,(Bi + 2)\,T = 0$$

It adds the conduction from the left and right neighbours (half height each), from the node above, and the convection h Dx (T_inf - T).

Hand solution: the eight equations

A. The eight node equations (Bi = 2.5, 2 Bi T_inf = 1500, 2(Bi+2) = 9)

```
node 1:   4 T1 - T2 - T3                  = 1000     (500 + 500 above and left)
node 2:  -2 T1 + 4 T2 - T4                =  500     (symmetry line, 500 above)
node 3:  -T1 + 4 T3 - T4 - T5             =  500     (500 on the left)
node 4:  -T2 - 2 T3 + 4 T4 - T6           =    0     (symmetry line)
node 5:  -T3 + 4 T5 - T6 - T7             =  500     (500 on the left)
node 6:  -T4 - 2 T5 + 4 T6 - T8           =    0     (symmetry line)
node 7:  -2 T5 + 9 T7 - T8                = 2000     (surface; 500 + 1500)
node 8:  -2 T6 - 2 T7 + 9 T8              = 1500     (surface, symmetry line)
```



B. Gaussian elimination: forward elimination gives an upper triangular system

```
R1:  4 T1 - 1 T2 - 1 T3                                    = 1000.00
R2:      3.5000 T2 - 0.5000 T3 - 1 T4                        = 1000.00
R3:          3.7143 T3 - 1.0714 T4 - 1 T5                    =  821.43
R4:              3.0962 T4 - 0.5769 T5 - 1 T6                =  759.62
R5:                  3.6770 T5 - 1.0932 T6 - 1 T7            =  791.93
R6:                      3.0270 T6 - 0.5946 T7 - 1 T8        =  716.22
R7:                          8.3393 T7 - 1.1964 T8           = 2571.43
R8:                                  7.9960 T8               = 2711.05
```

C. Back substitution (from R8 upwards)

```
T8 = 2711.05 / 7.9960                      = 339.05 K
T7 = (2571.43 + 1.1964 T8) / 8.3393        = 356.99 K
T6 = (716.22 + 0.5946 T7 + T8) / 3.0270    = 418.74 K
T5 = (791.93 + 1.0932 T6 + T7) / 3.6770    = 436.95 K
T4 = (759.62 + 0.5769 T5 + T6) / 3.0962    = 462.01 K
T3 = (821.43 + 1.0714 T4 + T5) / 3.7143    = 472.07 K
T2 = (1000 + 0.5 T3 + T4) / 3.5            = 485.15 K
T1 = (1000 + T2 + T3) / 4                  = 489.30 K
```

Hand solution: iteration, heat rate and energy balance

D. Gauss-Seidel iteration, starting from 400 K at all nodes (order 1 to 8)

iter	T1	T2	T3	T4	T5	T6	T7	T8
0	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00
1	450.00	450.00	437.50	431.25	434.38	425.00	363.19	341.82
2	471.88	468.75	459.38	453.12	436.89	417.18	357.29	338.77
3	482.03	479.30	468.01	458.13	435.62	417.03	356.67	338.60
4	486.83	482.95	470.14	460.07	435.96	417.65	356.72	338.75
5	488.27	484.15	471.08	460.99	436.36	418.12	356.83	338.88
13	489.30	485.15	472.07	462.01	436.95	418.74	356.99	339.05

Each node uses the newest values. The iteration stops when no temperature changes by more than 0.01 K: 13 iterations from 400 K (16 from 300 K). Gaussian elimination and Gauss-Seidel give the same temperatures.

E. Heat rate to the air stream (Task 5): trapezoidal rule on the bottom surface

```
q' = h Dx [ (500-300)/2 + (T7-300) + (T8-300) + (T7-300) + (500-300)/2 ]
   = 10 x 0.25 x [ 100 + 56.99 + 39.05 + 56.99 + 100 ]
   = 882.6 W/m
```

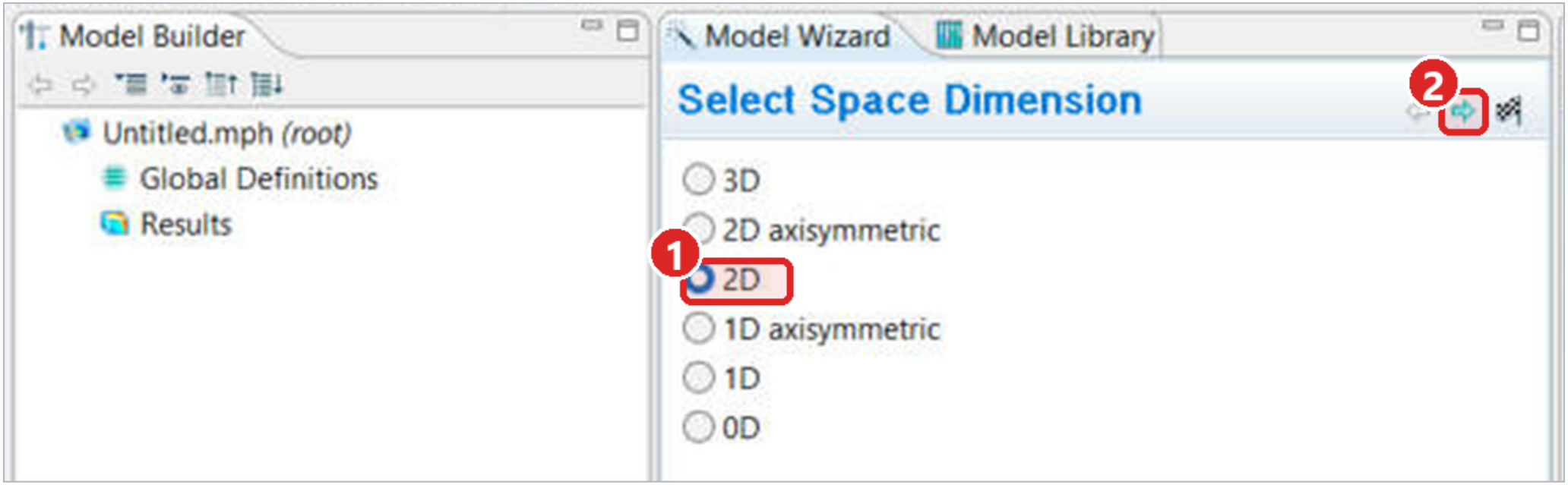
F. Energy balance (Task 6): heat conducted in = heat convected out

Item	W/m
conduction into the nodes 1 to 8 from the hot surfaces (top, left, right, and the two bottom corners)	382.6
= convection from the nodes 7, 8 and the mirror of 7: h Dx [2 (T7-300) + (T8-300)]	382.6
convection from the two corner cells (each h Dx/2 (500-300) = 250), supplied by the hot corner nodes	500.0
total heat rate q' to the air	882.6

The balance closes exactly (382.6 + 500.0 = 882.6): summing the control-volume equations of all nodes gives it. But 500 of the 882.6 W/m come from the two corner cells, whose temperature is fixed at 500 K and whose area is assigned by the trapezoidal rule: the heat rate is much more sensitive to the grid than the temperatures (results page).

Step 1 - Start a 2D model

Start COMSOL Multiphysics 4.3a. The Model Wizard opens (if not: File > New).

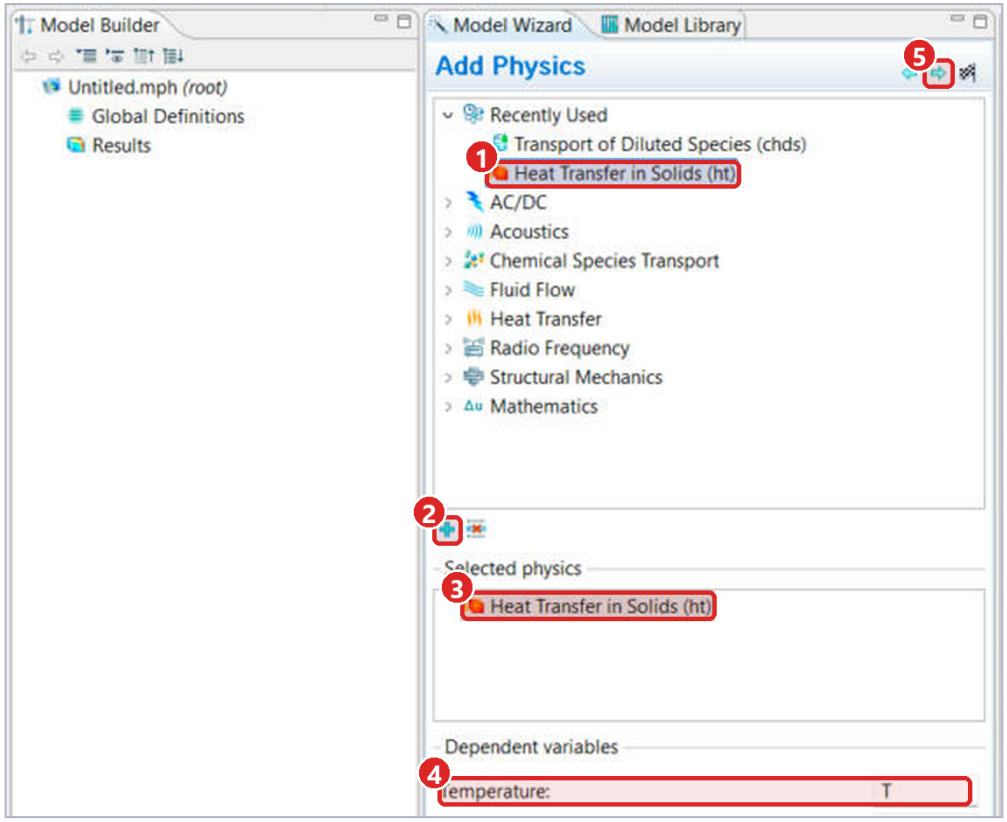


1. Select 2D. The column is long and nothing changes along its length (Assumption 2), so a cross-section in x and y is enough.
2. Click the Next arrow.

Why?
A long column with uniform conditions along its length is a 2D problem: the heat flows only in the plane of the cross-section.

Step 2 - Add Heat Transfer in Solids

Heat conduction in the brick is described by the Heat Transfer in Solids interface (physics name ht).

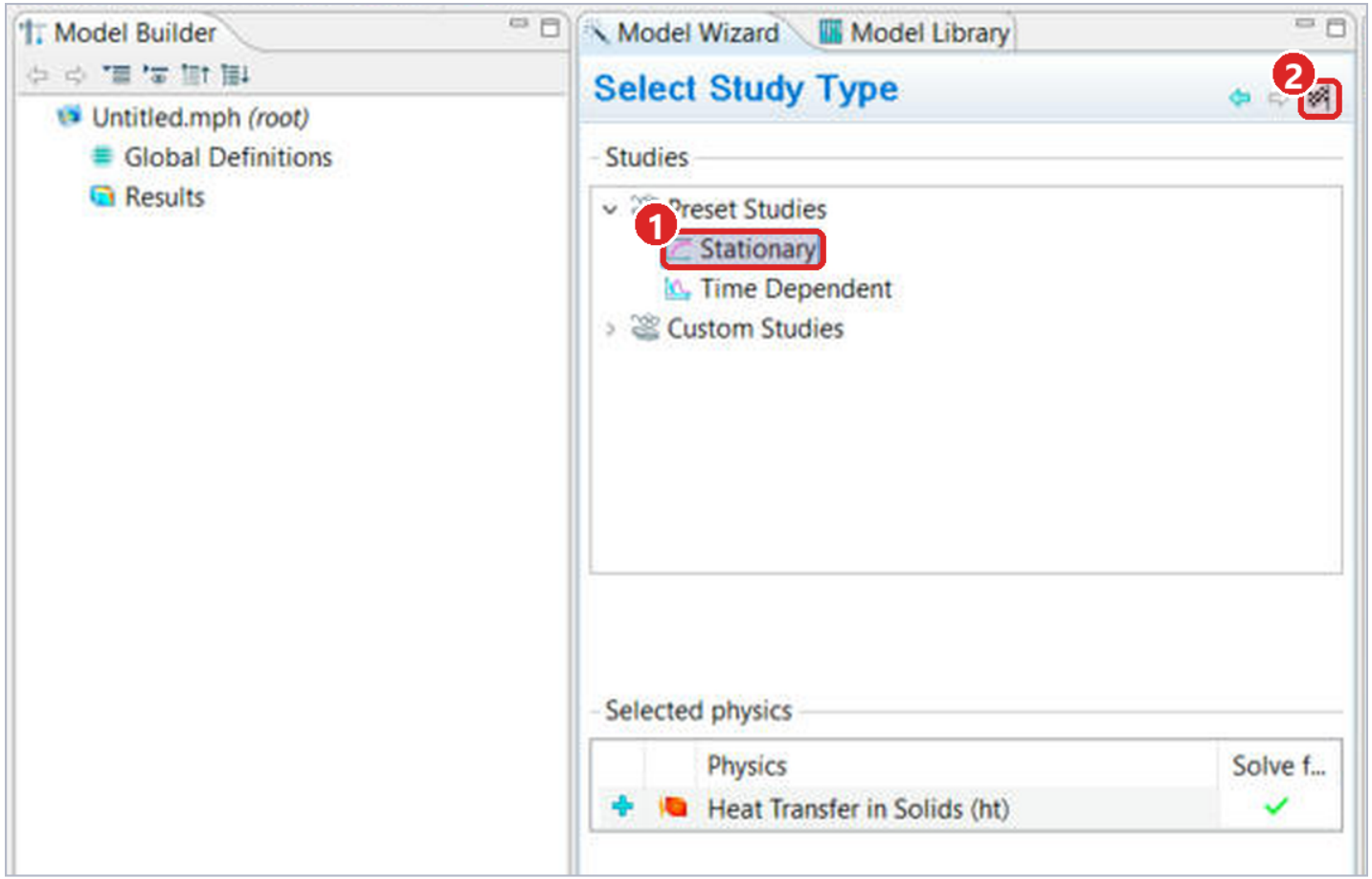


1. Select Heat Transfer in Solids (ht). It is in the folder Heat Transfer; it may also be listed under Recently Used.
2. Click the blue + (Add Selected) button.
3. The interface appears under Selected physics.
4. The dependent variable is the temperature T (in kelvin: the problem is in kelvin as well).
5. Click Next.

Why?
COMSOL solves rho Cp dT/dt + div(-k grad T) = Q. In steady state (no time derivative) with Q = 0 and constant k this is the Laplace equation d2T/dx2 + d2T/dy2 = 0 of Task 1.

Step 3 - Choose a stationary study

Steady-state operation means the temperatures no longer change with time.

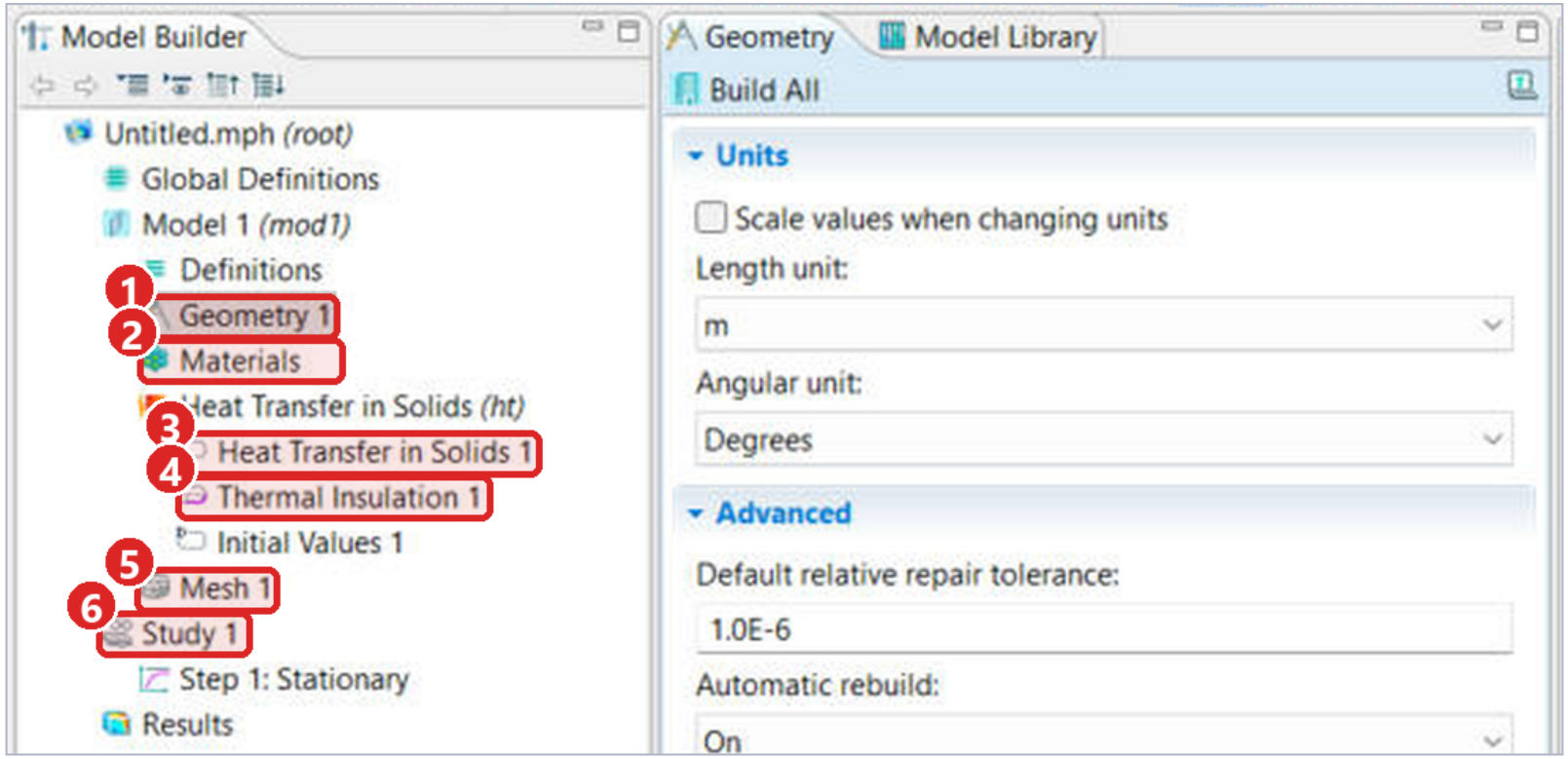


1. Select Stationary (under Preset Studies).
2. Click Finish (the chequered-flag icon).

Why?
A stationary study solves the whole system of node equations A T = b at once. The finite difference method does the same with Gaussian elimination or Gauss-Seidel iteration (Task 4).

Step 4 - Tour of the model tree

After Finish the Model Builder holds the model outline. Note what COMSOL has already created for us.

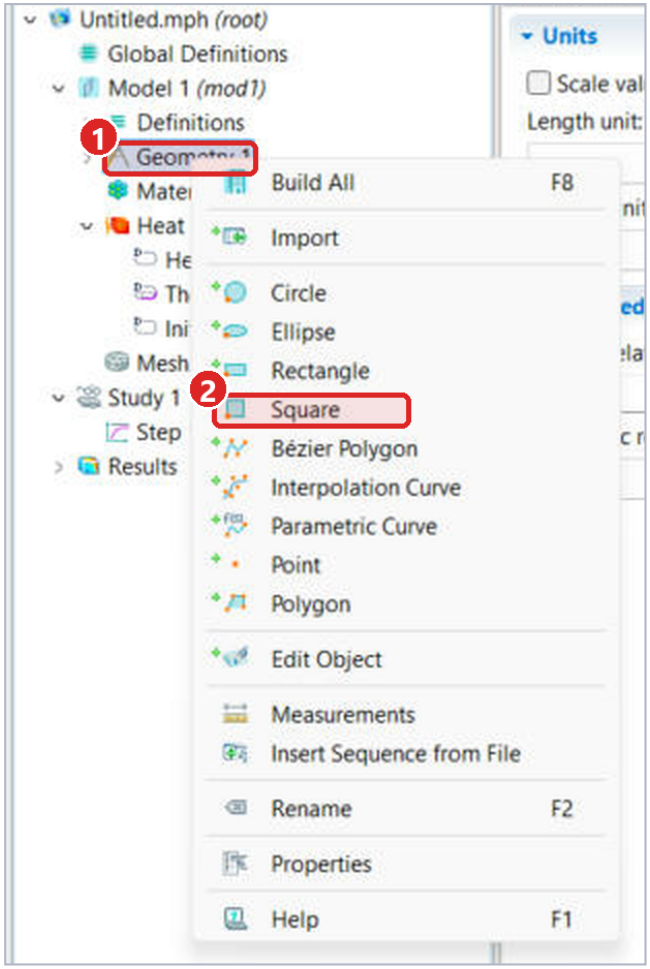


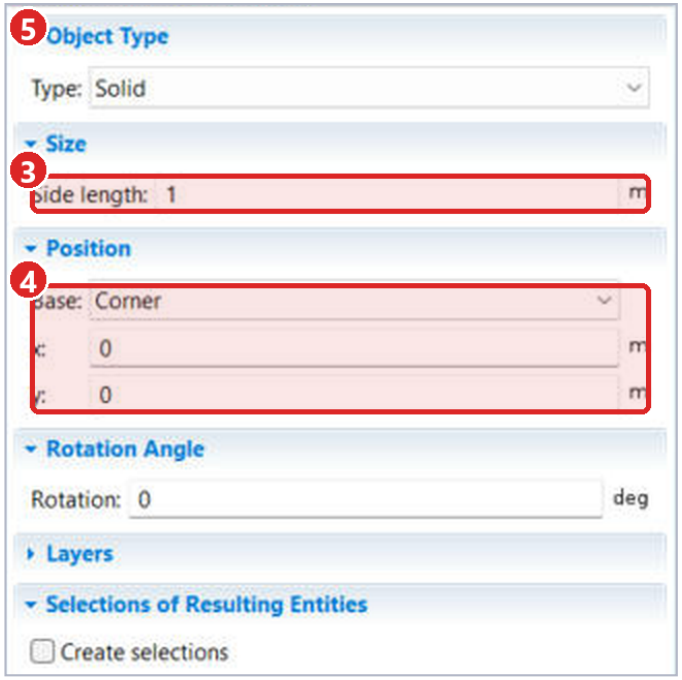
1. Geometry 1: the cross-section of the column (length unit: metre).
2. Materials: we must supply the thermal conductivity k = 1 W/(m K) of the fireclay brick.
3. Heat Transfer in Solids 1: the equation inside the brick.
4. Thermal Insulation 1: the default on every boundary. We will override it on all four edges: three Temperature conditions and one Convective Cooling condition.
5. Mesh 1: the grid (the problem uses Delta x = Delta y = 0.25 m).
6. Study 1 with Step 1: Stationary.

Why?
COMSOL will not use the symmetry line of the problem: it solves the whole 1 m x 1 m square. The symmetry of the result (T(x, y) = T(1 - x, y)) is then a free check of the solution.

Step 5 - Build the geometry: a 1 m square

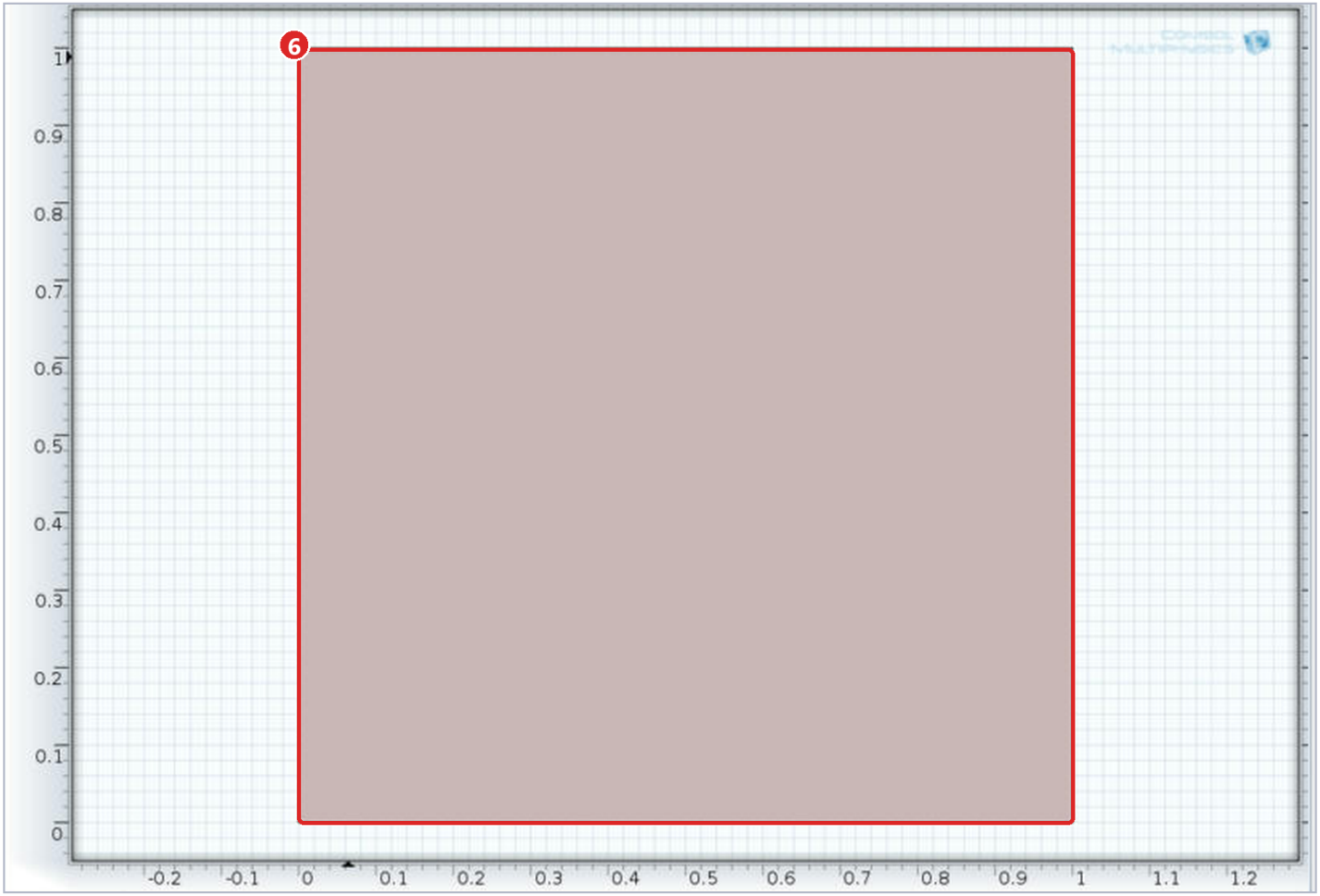
The cross-section of the column is 1 m by 1 m.





1. Right-click Geometry 1 (left-click it first if no menu appears).
2. Choose Square.
3. Side length = 1 m (the default).
4. Position: Base = Corner, x = 0, y = 0 (the defaults): 0 <= x <= 1 and 0 <= y <= 1.
5. Click Build Selected (or press F8).

Step 5 - Build the geometry: a 1 m square

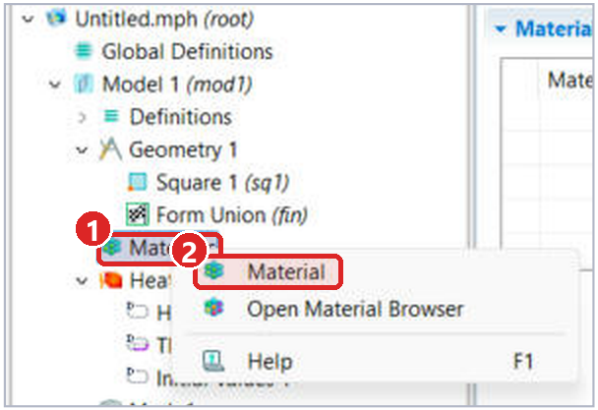


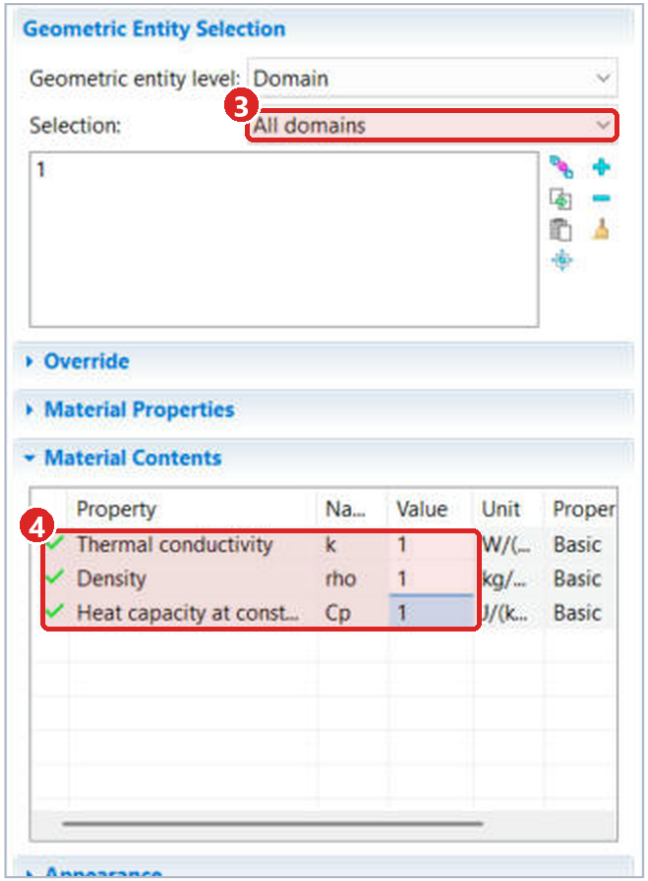
6 The square appears. The bottom edge (y = 0) will be the cooled surface, the other three edges are hot.

Why?
COMSOL numbers the four edges of this square: 1 = left, 2 = bottom, 3 = top, 4 = right. We will use these numbers when assigning the boundary conditions.

Step 6 - Define the material: fireclay brick

The problem gives k = 1 W/(m K) for the fireclay brick (at about 478 K).



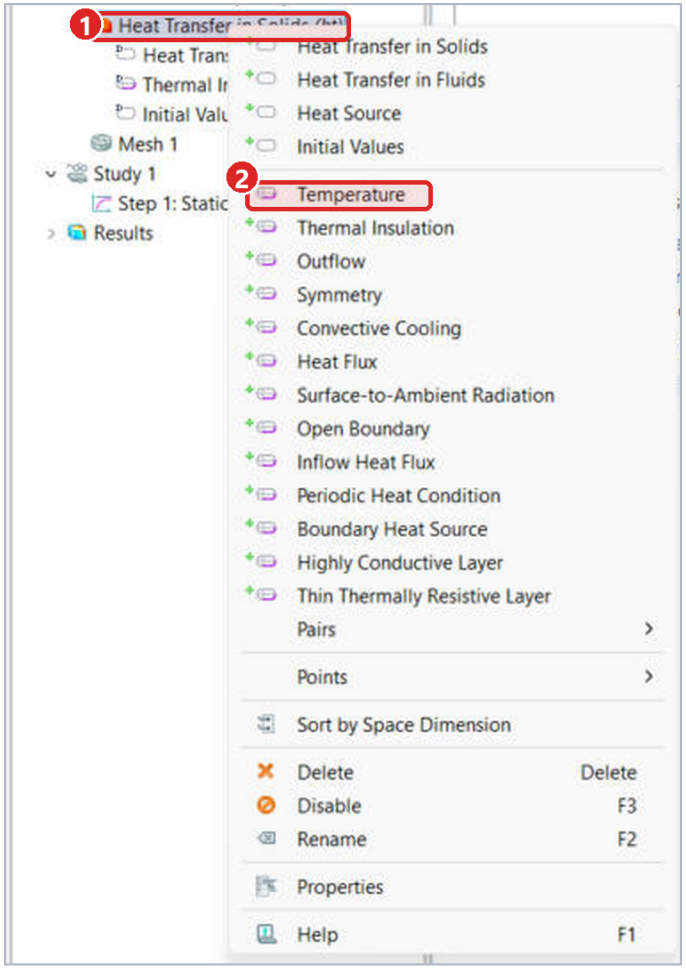


1. Right-click Materials (left-click it first).
2. Choose Material (a blank material).
3. Selection: All domains.
4. Enter k = 1 W/(m K). Density and heat capacity (rho = 1, Cp = 1) are not used in a stationary study, but COMSOL does not accept empty fields, so enter any value. All three rows then show a green tick.

Why?
Only k matters in this steady problem. With constant k the temperatures depend on k only through the Biot number Bi = h Dx / k = 10 x 0.25 / 1 = 2.5 of Task 1.

Step 7 - Hot surfaces: 500 K on three edges

The top, left and right surfaces are maintained at 500 K: a Dirichlet condition on edges 3, 1 and 4. One Temperature node can hold all three edges.



1. Left-click Heat Transfer in Solids (ht), then right-click it.
2. Choose Temperature.

Step 7 - Hot surfaces: 500 K on three edges



3 Selection list: 1, 3, 4 (left, top, right). For each edge: hover until it turns red, click it, press the + button (Add to Selection).

4 Temperature T0 = 500[K]. Click in the field, press Ctrl+A, type the value and press Enter.

5 The three selected edges are drawn in blue; the bottom edge is left out.

Why?
The temperature of the corner points is 500 K as well, because they belong to the hot edges (the problem has the same value at the corners of the cooled surface).

Step 8 - Cooled surface: convection to the air

The bottom surface exchanges heat with the air stream: -k dT/dn = h (T - T_inf), with h = 10 W/(m^2 K) and T_inf = 300 K.

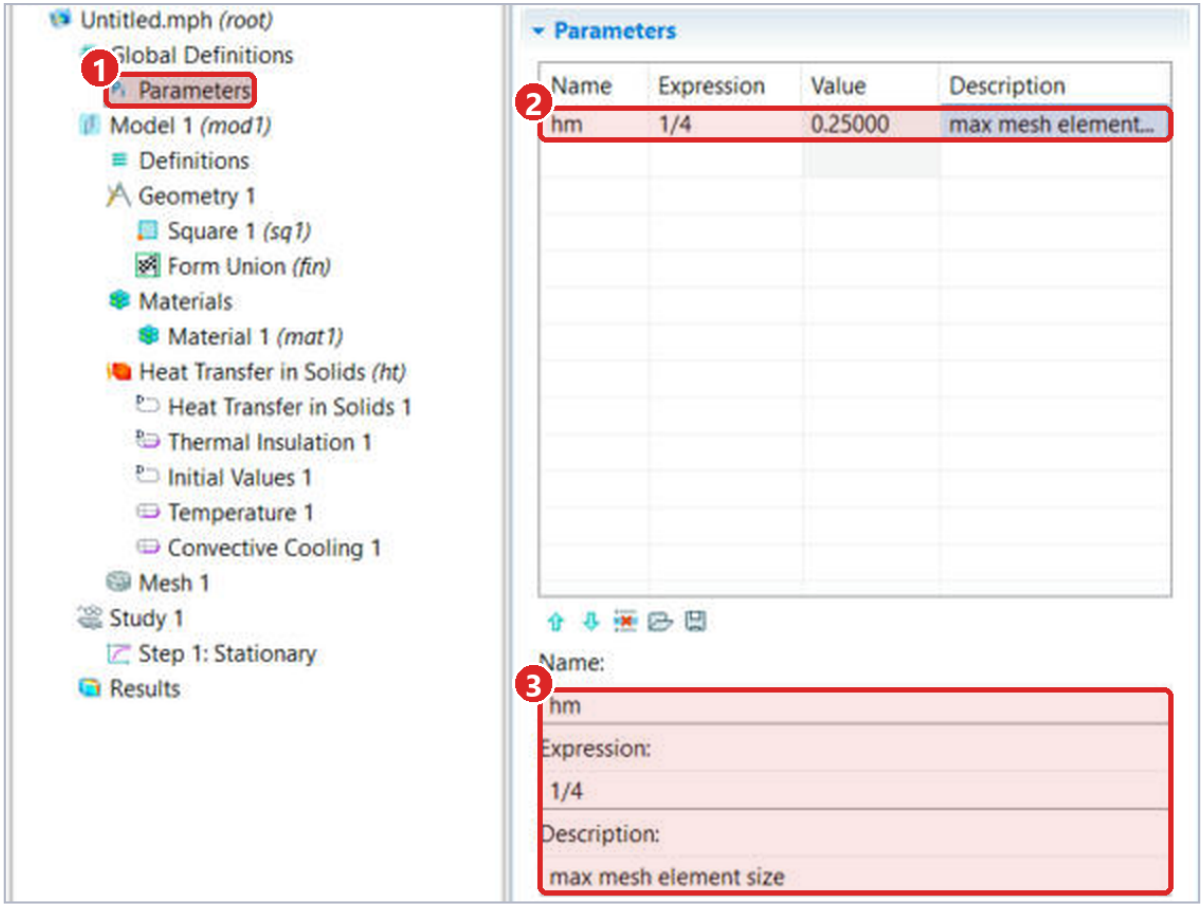


1. Right-click Heat Transfer in Solids (ht) > Convective Cooling. The selection list must contain 2 (the bottom edge): hover over it until it turns red, click it, press +.
2. Heat transfer coefficient: User defined, h = 10 W/(m^2 K).
3. External temperature T_ext = 300[K].
4. The bottom edge is drawn in blue.

Why?
This is the convective (Robin) boundary condition. The finite difference method includes it with an energy balance on a half control volume at the surface; COMSOL adds the same physics as a boundary integral. Radiation is neglected (Assumption).

Step 9 - Make the mesh size a parameter

We want to repeat the calculation on finer meshes (Task 6). A parameter hm for the element size makes that easy.

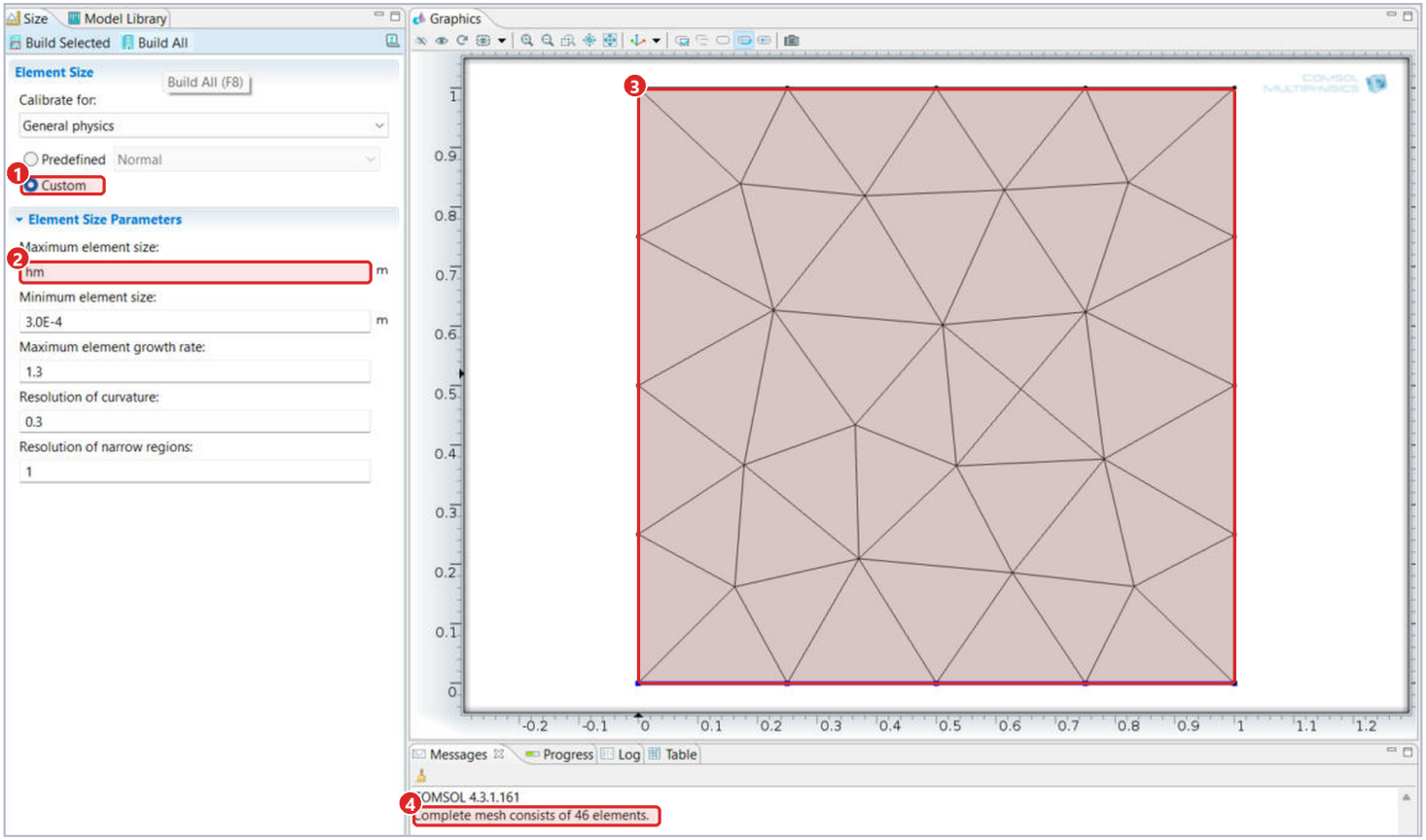


1. Right-click Global Definitions > Parameters (left-click Global Definitions first).
2. First row: Name hm, Expression 1/4, Description max mesh element size. Type the Name, press Tab, type the Expression, press Tab, type the Description, press Enter.
3. The fields below the table show the same row: hm = 1/4 (value 0.25).

Why?
hm = 1/4 m is the grid spacing of the problem. We will sweep it over 1/4, 1/8, 1/16, 1/32 and 1/64.

Step 10 - Create the mesh with maximum size hm

COMSOL meshes the square with triangles; we limit the element size with the parameter hm.



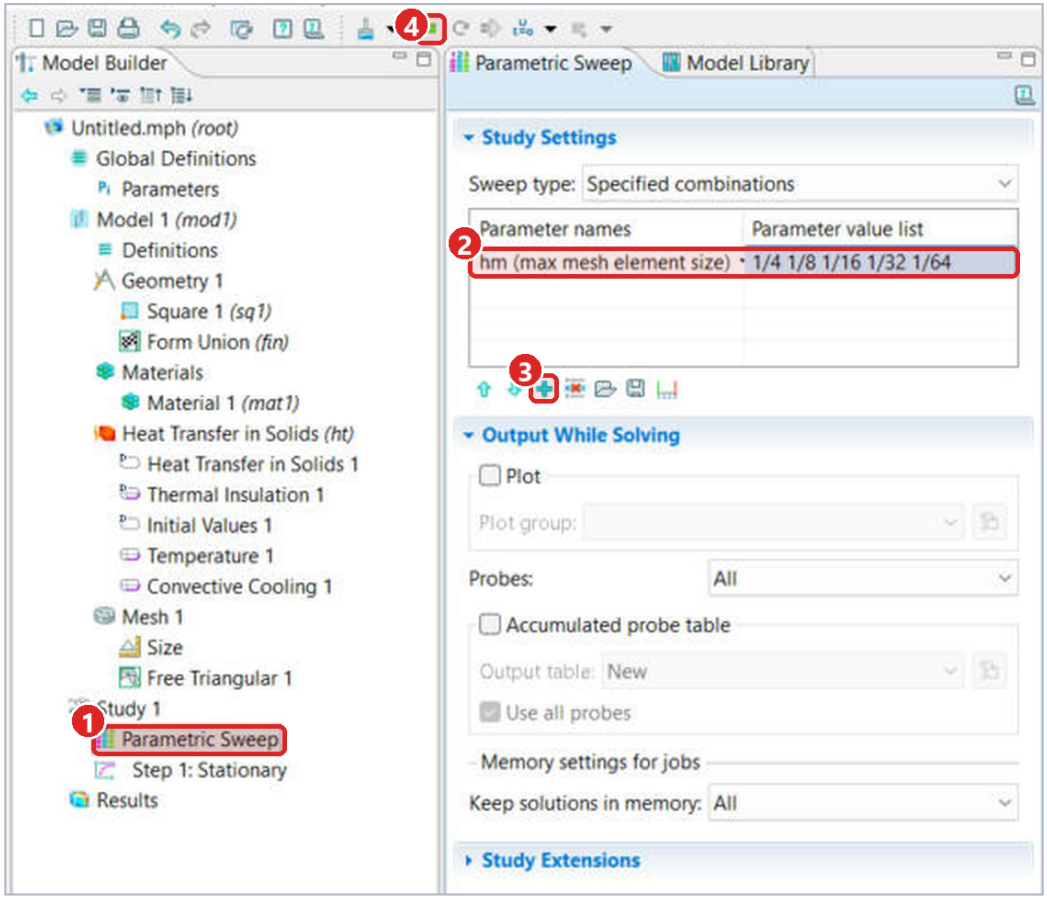
1. Click Mesh 1, set Sequence type to User-controlled mesh, then click the node Size and select Custom.
2. Maximum element size = hm. Click Build All (F8).
3. The mesh for hm = 0.25: about 4 elements along every edge.
4. Messages: Complete mesh consists of 46 elements.

Why?

With hm = 0.25 each edge has four elements, similar to the 4 x 4 nodes of the problem. COMSOL uses triangles with quadratic shape functions, so its solution on a mesh of this size is not the same as the finite difference solution on the square grid.

Step 11 - Add a parametric sweep over hm

A parametric sweep solves the model once per value of hm and rebuilds the mesh each time.

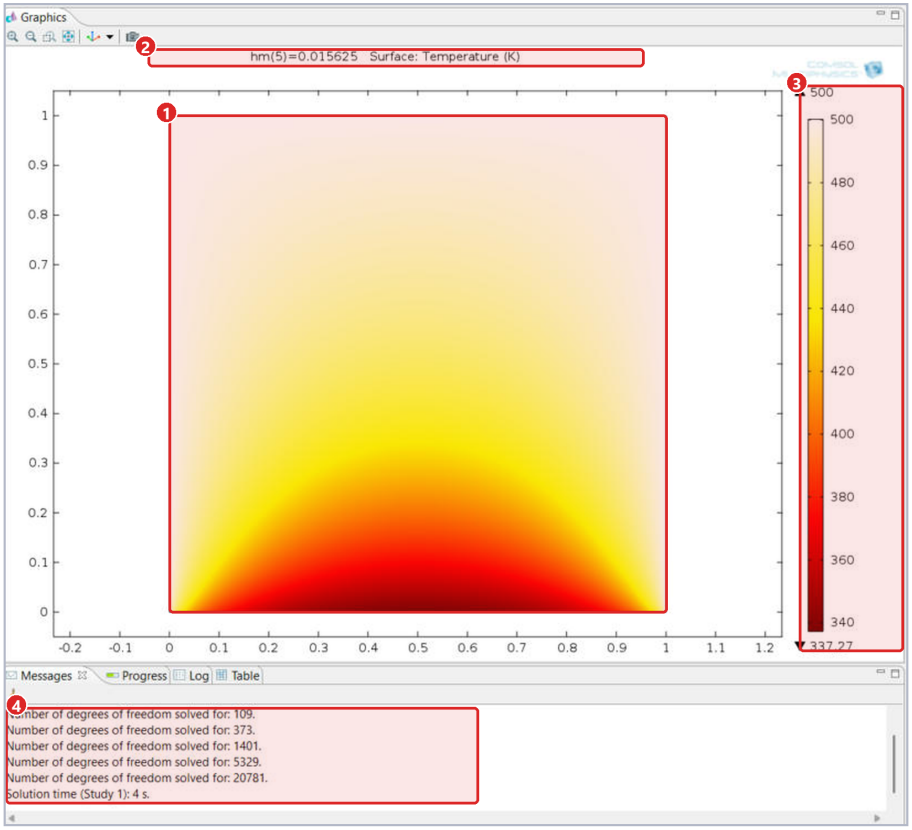


1. Right-click Study 1 > Parametric Sweep (left-click Study 1 first).
2. Parameter names = hm, Parameter value list = 1/4 1/8 1/16 1/32 1/64 (values separated by spaces).
3. The + button adds a row and offers the parameters of Global Definitions.
4. Click Compute (the green = in the toolbar, or press F8).

Why?
The finest mesh (hm = 1/64) has 20 781 degrees of freedom; the whole sweep takes about 4 s.

Step 12 - Compute and look at the temperature field

After Compute, COMSOL plots the temperature of the last parameter value (hm = 1/64).



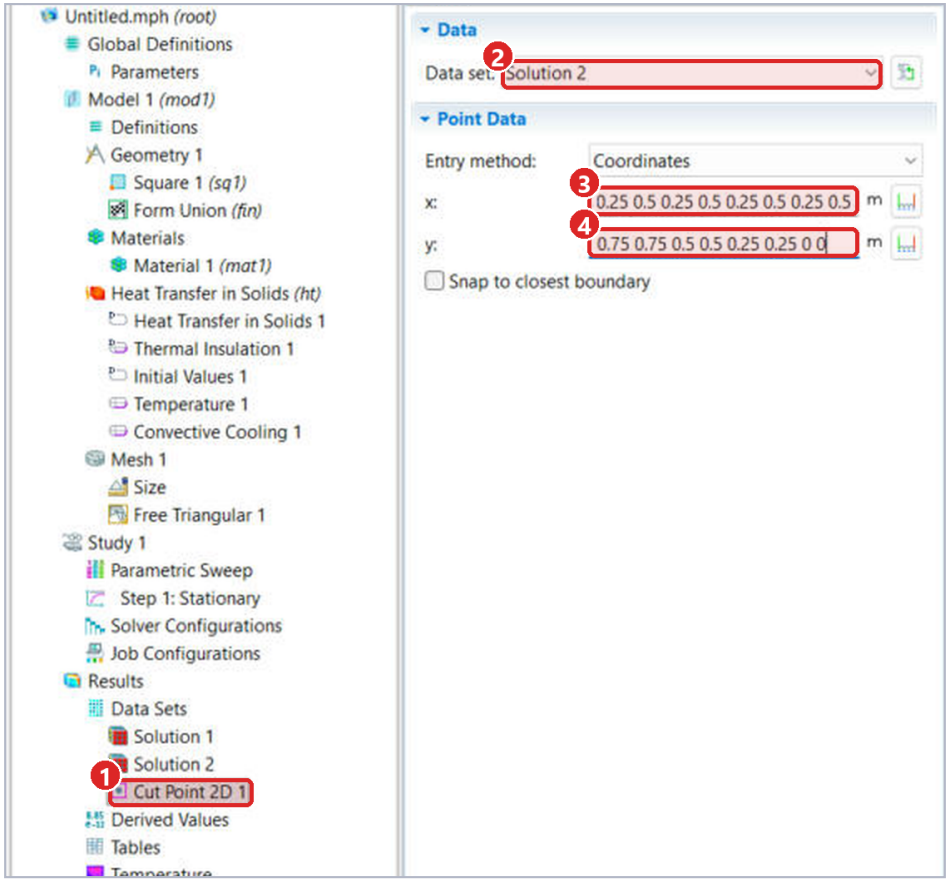
1. The temperature surface plot: about 500 K nearly everywhere, cooling towards the bottom surface.
2. Title: hm(5) = 0.015625 = 1/64, the 5th and finest mesh.
3. Colour legend in kelvin: from 337.27 K (coldest, on the bottom surface) to 500 K.
4. Messages: degrees of freedom 109, 373, 1401, 5329, 20781; Solution time (Study 1): 4 s.

Why?

The coldest point is the middle of the cooled surface (337.27 K), as in the finite difference solution (T8 = 339 K on the 4 x 4 grid). The temperature is symmetric about the vertical centre line.

Step 13 - Read the temperature at the eight nodes

The eight unknowns of the finite difference problem are at x = 0.25 and 0.5, y = 0.75, 0.5, 0.25 and 0. We create a Cut Point 2D data set with these eight points.

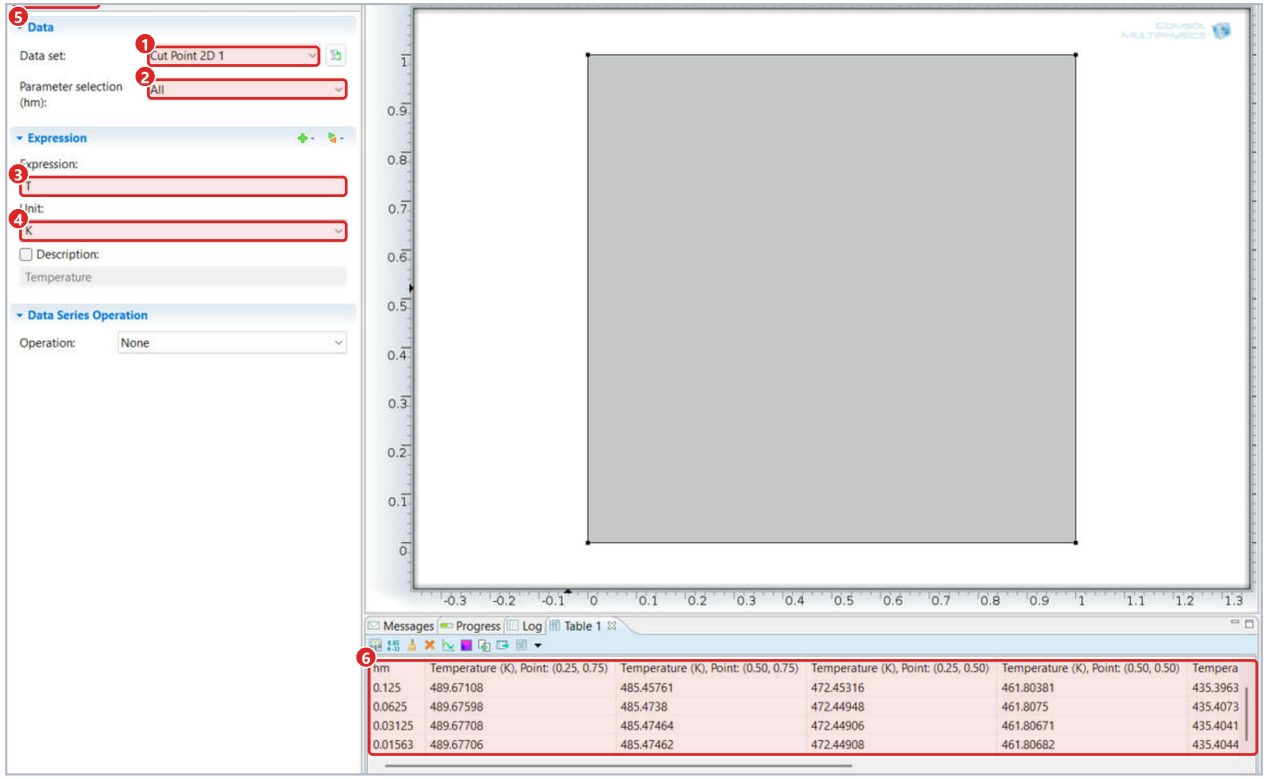


1. Right-click Results > Data Sets > Cut Point 2D. The node Cut Point 2D 1 appears.
2. Data set: Solution 2 (the parametric solutions).
3. x: 0.25 0.5 0.25 0.5 0.25 0.5 0.25 0.5
4. y: 0.75 0.75 0.5 0.5 0.25 0.25 0 0. Points in the order of the nodes 1 to 8 of the problem.

Why?
Point k uses the k-th x and the k-th y. The points with y = 0 lie on the cooled surface; COMSOL evaluates them there without any special setting.

Step 14 - Evaluate the temperature (point evaluation)

A Point Evaluation puts the temperature at the eight points into a table, one row per mesh.



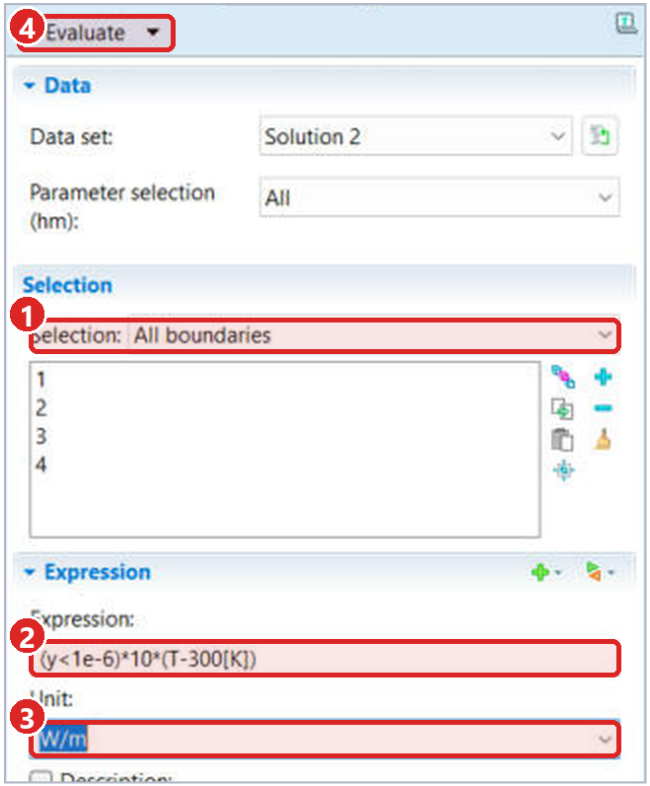
1. Right-click Results > Derived Values > Point Evaluation. Data set: Cut Point 2D 1.
2. Parameter selection (hm): All, so every mesh gets a row.
3. Expression: T.
4. Unit: K (kelvin, as in the problem).
5. Click Evaluate.
6. Table 1 opens at the bottom: one row per hm (the first row, hm = 0.25, is scrolled out of view), one column per point. Double-click the tab to enlarge it; the export button saves it as a text file.

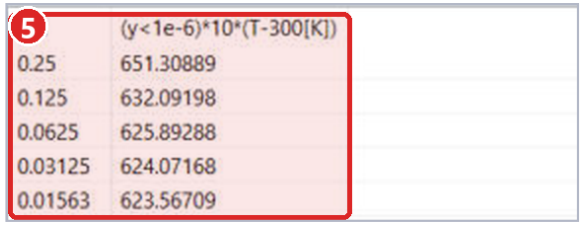
Why?

On the finest mesh the eight values are 489.68, 485.47, 472.45, 461.81, 435.40, 416.15, 348.36 and 337.27 K. They agree with the finite difference solution on a 256 x 256 grid to better than 0.001 K, see the results page.

Step 15 - Heat rate to the air stream

The heat rate per unit length is the integral of the convective heat flux h (T - T_inf) over the bottom surface. We evaluate it with a line integral.





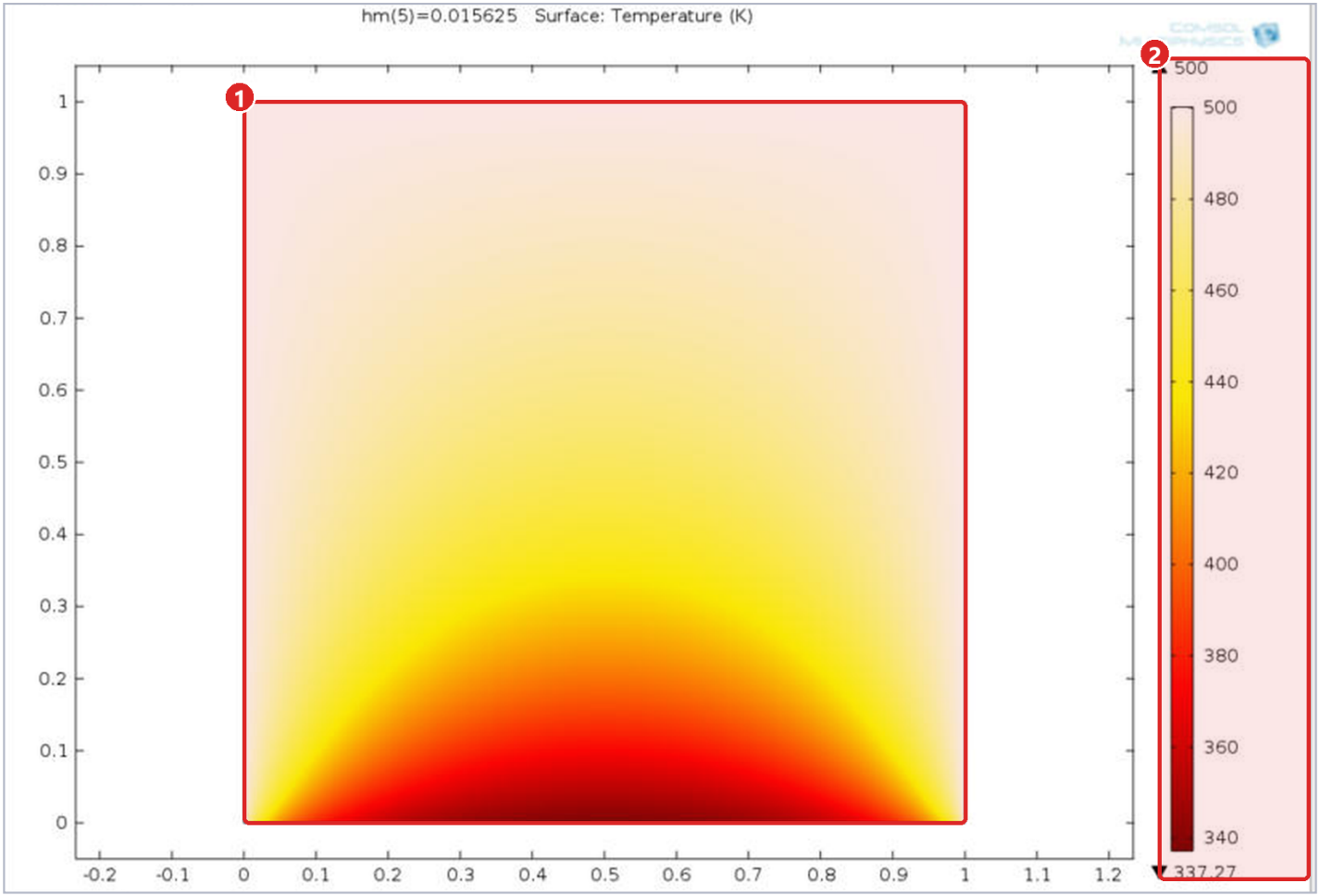
1. Right-click Derived Values > Integration > Line Integration. Data set: Solution 2. Selection: All boundaries.
2. Expression: (y<1e-6)*10*(T-300[K]) : the convective flux h (T - T_inf) on the bottom edge (the logical factor y<1e-6 is 1 only there and 0 on the other edges).
3. Unit: W/m (heat per unit length of the column). COMSOL may display the field empty after Evaluate: the result is in W/m.
4. Click Evaluate.
5. Table 2: the heat rate q' in W/m for the five meshes.

Why?

q' = 651.3, 632.1, 625.9, 624.1 and 623.6 W/m for hm = 1/4 ... 1/64. The value is converging to about 623.4 W/m. The finite difference solution on the 4 x 4 grid gives q' = 882.6 W/m, 41 % higher: see the results page for the reason.

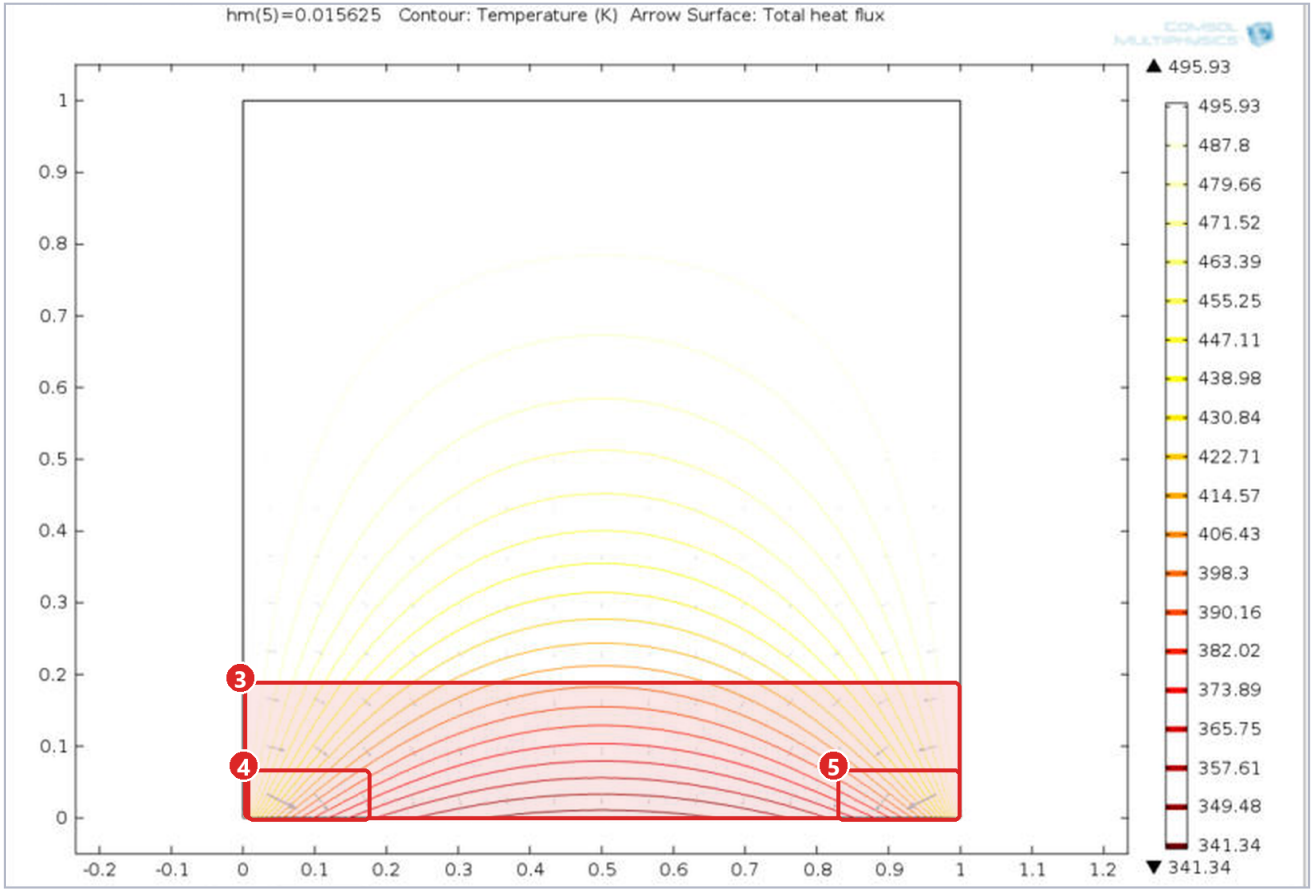
Step 16 - Plot the temperature field and isotherms

Plots help to see whether the result is physically reasonable.



1. Plot group Temperature (ht): the column is almost isothermal (500 K) except near the cooled bottom surface.
2. Colour legend in kelvin.

Step 16 - Plot the temperature field and isotherms



3 Plot group Isothermal contours (with arrows for the heat flux): the isotherms are arches over the cooled surface; the heat flows from the hot walls towards the bottom.

4 Bottom left corner: the 500 K side wall meets the cooled surface; the isotherms converge here.

5 Bottom right corner: mirror image (symmetry).

Why?
The temperature gradient, and therefore the heat flux, is largest near the two bottom corners. More than half of the heat rate computed on the coarse 4 x 4 grid (500 of 882.6 W/m) comes from the two corner cells, which is why that grid is good for temperatures but poor for the heat rate (see the results page).

A. Temperatures (K) at the eight nodes

Node	(x, y)	FDM 4 x 4	FDM 256 x 256	COMSOL hm=1/64	4 x 4 minus converged
1	(.25,.75)	489.30	489.677	489.677	-0.37
2	(.5,.75)	485.15	485.474	485.475	-0.32
3	(.25,.5)	472.07	472.449	472.449	-0.38
4	(.5,.5)	462.01	461.807	461.807	+0.20
5	(.25,.25)	436.95	435.405	435.404	+1.54
6	(.5,.25)	418.74	416.153	416.152	+2.59
7	(.25,0)	356.99	348.363	348.363	+8.63
8	(.5,0)	339.05	337.274	337.274	+1.78

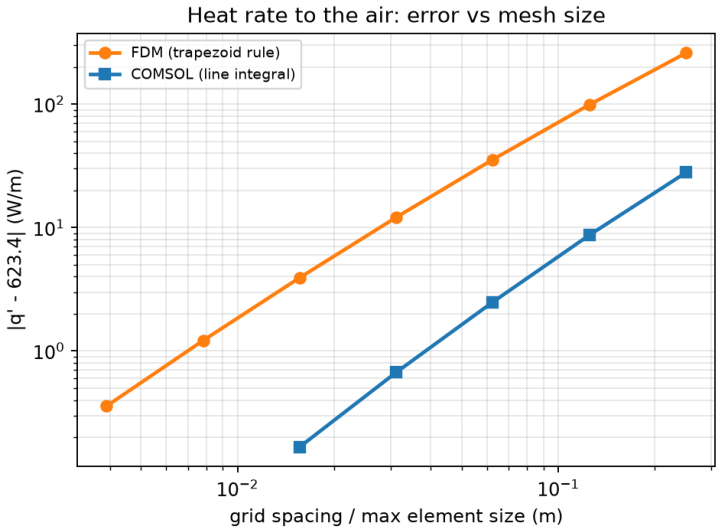
The error of the 4 x 4 grid grows towards the cooled surface: within 0.4 K at nodes 1 to 4, 1.5 K and 2.6 K at nodes 5 and 6, 1.8 K at node 8 and 8.6 K (too warm) at node 7, the surface node next to the corner. COMSOL on its finest mesh and the 256 x 256 finite difference grid agree to better than 0.001 K at every node.

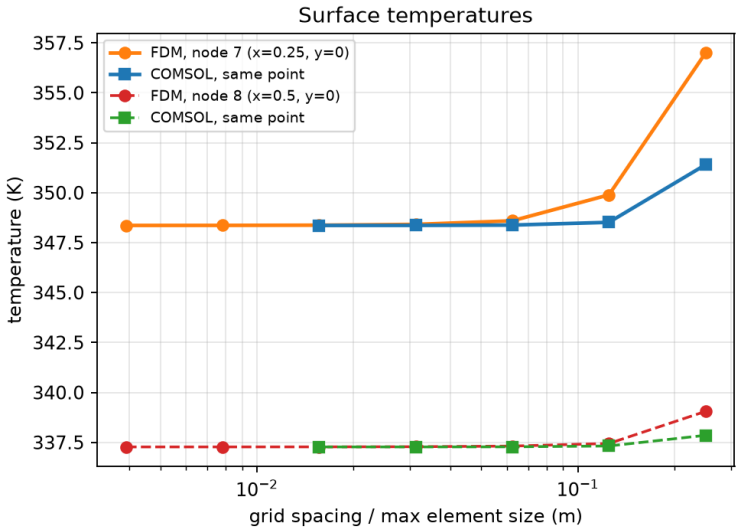
B. Heat rate q' (W/m) and surface temperatures: finite difference grid study (Task 6)

N (cells per side)	Dx (m)	T7 (K)	T8 (K)	q' (W/m)	q' minus 623.4
4	0.25000	356.99	339.05	882.6	+259.2
8	0.12500	349.89	337.45	722.3	+98.9
16	0.06250	348.59	337.32	659.0	+35.6
32	0.03125	348.42	337.28	635.5	+12.1
64	0.01562	348.38	337.28	627.3	+3.9
128	0.00781	348.37	337.27	624.6	+1.2
256	0.00391	348.36	337.27	623.8	+0.4

C. COMSOL mesh study (Steps 9 to 15)

hm (m)	degrees of freedom	T7 (K)	T8 (K)	q' (W/m)
0.25000	109	351.40	337.85	651.3
0.12500	373	348.52	337.32	632.1
0.06250	1401	348.38	337.28	625.9
0.03125	5329	348.36	337.27	624.1
0.01562	20781	348.36	337.27	623.6





Why the heat rate converges more slowly than the temperatures

What the grid study shows

- The temperatures at nodes 1 to 4 (the upper half of the column) are within 0.4 K of the converged solution on the 4 x 4 grid; the error grows towards the cooled surface (2.6 K at node 6, 8.6 K at node 7). The overall temperature distribution is captured well, the values near the cooled surface are not.

- The heat rate is not: q' = 882.6 W/m on the 4 x 4 grid, 722 W/m on 8 x 8, 659 W/m on 16 x 16 and about 623.4 W/m converged. The 4 x 4 grid over-predicts it by 41 %.

- The reason: near the bottom corners the temperature changes quickly from 500 K (side wall) to about 348 K (cooled surface). A coarse grid represents this region badly, and the trapezoidal rule gives the two corner cells (500 K) a weight of half a cell each; they contribute 500 of the 882.6 W/m.

- COMSOL converges faster at the same mesh size (651 W/m at hm = 1/4) because it uses quadratic elements and integrates the flux over the whole edge. Both methods converge to the same heat rate, about 623.4 W/m (an extrapolated value, consistent with both series); the error falls by a factor of about 3 per halving of the mesh size, between first and second order.

- Rule of thumb: check a result that depends on a gradient (heat flux, heat rate) with at least one refined grid; temperatures converge faster than their derivatives. The heat rate and the surface temperatures are the quantities that need the finer grid.

Troubleshooting and tips

The right-click menu does not appear.

Left-click the node once to select it, then right-click again.

A selection list stays empty or contains the wrong edge.

Hover the cursor over the edge until it turns red, click it, then press the + button. If the list shows an edge you did not want, select it in the list and press the minus button. Edges: 1 = left, 2 = bottom, 3 = top, 4 = right.

Typing in a field added to the old text.

Click in the field, press Ctrl+A to select everything, then type the new value.

The temperatures look wrong by about 273.

COMSOL works in kelvin. The problem is in kelvin as well: type 500[K]. If you type 500 without a unit COMSOL assumes kelvin; 500[degC] would be 773 K.

The bottom edge is not cooled (the whole plate is at 500 K).

The Convective Cooling node must contain edge 2 in its selection list, and h must be 10 (the default is 0).

The heat rate differs from the table.

Check the expression (y<1e-6)*10*(T-300[K]): the factor y<1e-6 selects the bottom edge. Use the data set Solution 2 and Selection: All boundaries.

The Point Evaluation table is empty.

Evaluate was not pressed, or the data set is not Cut Point 2D 1.

COMSOL asks for a license.

Start COMSOL with the shortcut your installation provides. On campus the computer must be on the university network or VPN.

Save your work.

File > Save As, choose a folder you can write to. Compare with the supplied file brick_column_comsol.mph.